

Social Text Sentiment

- From a Dictionary to a Language Model -

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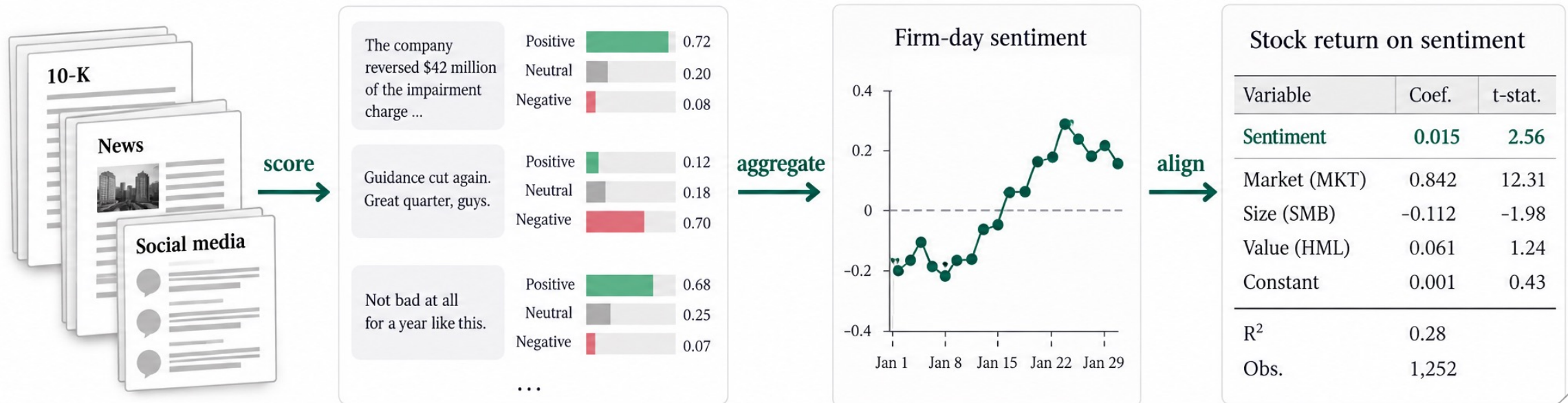
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Text Sentiment as Quantified Alternative Data

Counting words says how much was written, not whether the news was good or bad.

- Sentiment analysis is the part of natural language processing that asks that question.
- Three generations of methods: a word list, a fine-tuned encoder, and a prompted language model.
- Each one returns a score per sentence, which is then aggregated into one number per firm per day.



One Sentence, Scored Two Ways

A word list scores this sentence as negative. Reading it gives the opposite answer.

"The Company reversed \$42 million of the impairment charge recorded in the prior year, and settled the outstanding litigation at no cost."

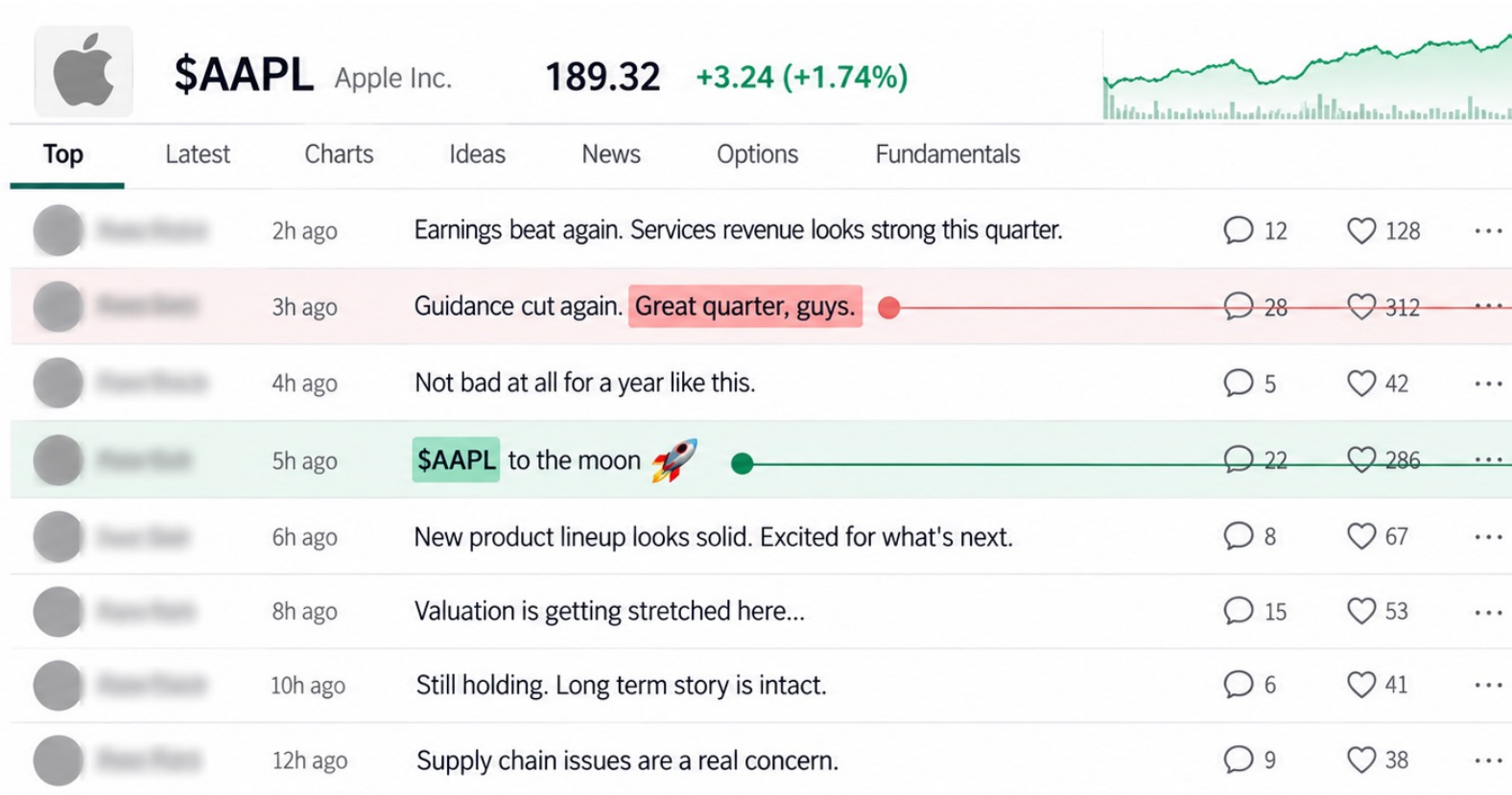
Method	What it counts	Verdict
Word-list counting	impairment, charge, litigation, cost: four negatives, no positives	Strongly negative
Reading the sentence	A reversal adds income; a settlement at no cost removes a liability	Positive

- The list is not broken. It cannot see that reversed and at no cost invert all four words.



The Same Method on a Social Post

The same word-list method, applied to posts from a stock discussion board.



Sarcastic post

"Guidance cut again. Great quarter, guys."

A word list counts "great" as positive, but the post is sarcastic, so the true label is **negative**.

Cashtag-only post

"**\$AAPL to the moon** 🚀"

No listed word is matched, but the post is **bullish**.

The Case for a Finance-Specific Dictionary

Loughran and McDonald (2011), When Is a Liability Not a Liability?, Journal of Finance 66(1).

About three-quarters of the general-English negative hits in a 10-K are not negative there.

Word, on a general negative list	What it usually means in a 10-K	Negative?
tax	A line item every company has	No
cost	Cost of goods sold, an accounting category	No
liability	The right-hand side of a balance sheet	No
impairment	A write-down of an asset's carrying value	Yes

- The response was a purpose-built finance word list: free, deterministic, instantaneous, and auditable word by word.



Quantifying a Document With a Word List

How the first generation produced a number, in one line of arithmetic.

$$\text{score}(d) = \frac{n_{\text{pos}}(d) - n_{\text{neg}}(d)}{n_{\text{words}}(d)}$$

- Every hit is worth one, and the divisor keeps long and short documents comparable.
- Averaging over a firm's documents in a day gives the firm-day series.

Document sentence (from a filing)

The Company reversed \$42 million of the **impairment** **charge** and settled the **litigation** at no **cost**.

↓ ↓ ↓ ↓

negative negative negative negative

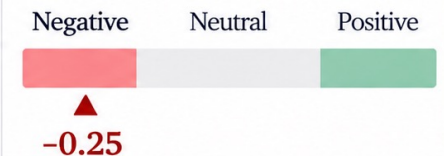
Word-list counting

Positive words	0
Negative words	4
Total words	16

Score calculation

$$\begin{aligned}\text{score}(d) &= \frac{n_{\text{pos}}(d) - n_{\text{neg}}(d)}{n_{\text{words}}(d)} \\ &= \frac{0 - 4}{16} = -0.25\end{aligned}$$

Document sentiment (word-list)



The Limits of a Dictionary Method

Five properties of a dictionary method, and what each one costs.

Property	Consequence
It scores a word, not a sentence	Word order is discarded entirely
A word has one score everywhere	Volatile is negative in a risk factor, neutral about a chemical
An unlisted word contributes nothing	Most filing sentences contain no listed word and default to neutral
It cannot represent negation	"Not without risk" and "substantial risk" score alike
Nothing in it was fitted	No weight was ever learned from a labeled outcome



Domain Adaptation of a General Language Model

The pretraining corpus decides what the representations are good at.

- The problem: a model that read little financial prose represents it poorly.
- The repair: continue the masked-token objective on financial text, then fine-tune.
- Unchanged: the architecture, the tokenizer and the vocabulary.

Domain adaptation is the same method applied to a different corpus.



FinBERT as the Repair for the Dictionary

FinBERT is a BERT trained further on financial text, then fine-tuned to label sentences.

- It exists to fix one thing: a general model represents financial prose poorly.
- What it returns is a probability vector per sentence, not a count of matched words.

What defines it is the corpus it was trained on and the labels it was fine-tuned on.

Word-list method (counts individual words)

The Company reversed \$42 million of the impairment charge and settled the litigation at no cost.

Positive words	0
Negative words	4
Total words	16



Document sentiment

(word-list score)

$$\text{score}(d) = \frac{n_{\text{pos}}(d) - n_{\text{neg}}(d)}{n_{\text{words}}(d)}$$
$$= \frac{0 - 4}{16} = -0.25$$

FinBERT (scores the whole sentence)

The Company reversed \$42 million of the impairment charge and settled the litigation at no cost.



FinBERT
(pretrained on
financial text,
fine-tuned for
sentiment)



Sentence sentiment (probabilities)

Positive		0.72
Neutral		0.18
Negative		0.10



Further Pretraining on a Financial Corpus

The same fill-in-the-blank objective, continued on a new corpus.

Stage	Corpus	Scale, as reported
General pretraining	Wikipedia and books	The BERT-base checkpoint as released
Further pretraining, Araci (2019)	Reuters TRC2 financial news	46,143 documents, over 29 million words
Further pretraining, Yang et al. (2020)	Filings, calls, analyst reports	About 4.9 billion tokens
Fine-tuning	Labeled financial sentences	A few thousand sentences

- The gain sits in the parameters, not in the tokenizer: EBITDA still splits into the same word pieces.



The Financial PhraseBank

The labeled dataset behind the first FinBERT.

Property	Value, as reported by the dataset
Sentences	4,845 English sentences drawn from financial news
Annotators	16 people with a background in financial markets
Labels	positive, neutral, negative, from an investor's point of view

Annotators who agreed	100%	At least 75%	At least 66%	At least 50%
Sentences retained	2,264	3,453	4,217	4,846

- One dataset filtered four ways, so a reported figure without its agreement level means little.

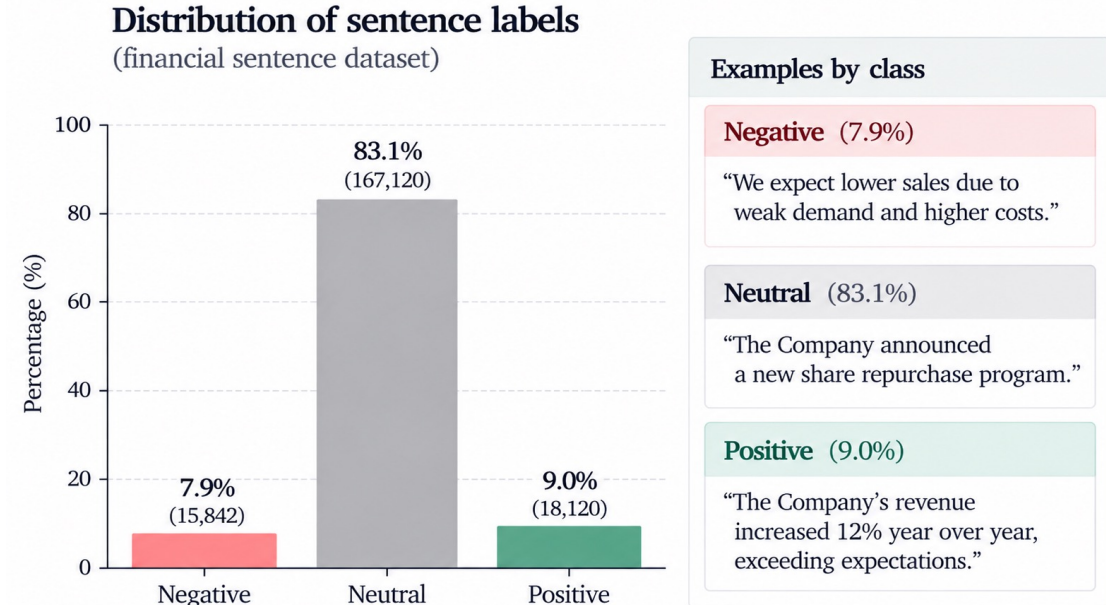


Three Classes and the Dominance of Neutral

Neutral is by far the largest of the three classes.

Consequence	The practical response
Answering neutral for everything already scores well	Report per-class recall, not accuracy alone
Boilerplate is genuinely neutral, and there is a lot of it	Score the exhibit that carries the news
Document averages are pulled toward zero	Consider a rule that excludes neutrals, and report both

- Most sentences report something without a direction:
a date, a legal formula, an unchanged figure.



The Procedure for Applying FinBERT

The procedure for running FinBERT on your own corpus, in six steps.

1. Acquire the documents, with an identifier and a timestamp.
2. Split every document into sentences.
3. Classify each sentence, keeping the three probabilities.
4. Aggregate to one score per document.
5. Aggregate again to one score per firm-day, under a stated lag rule.
6. Regress against a baseline, inside purged folds.

Steps 4 and 5 are choices you have to make, not defaults.



Step 2: Splitting One Exhibit

The five sentences carried through the rest of this session.

#	Sentence, as the splitter returned it
s1	Revenue for the quarter increased 8% to \$1.24 billion.
s2	Operating margin declined 210 basis points year over year.
s3	The Company reversed \$42 million of the impairment charge recorded in the prior year.
s4	This report contains forward-looking statements within the meaning of the Reform Act of 1995.
s5	The Board declared a quarterly dividend of \$0.24 per share, unchanged from the prior quarter.

- Two of the five carry the news, so a rule that treats all five equally has already chosen.



Step 3: The Five Sentences Scored

Illustrative output from a fine-tuned encoder such as FinBERT. Each row's three probabilities sum to one.

#	p(positive)	p(neutral)	p(negative)	Label	Signed score
s1	0.94	0.05	0.01	positive	+0.93
s2	0.02	0.06	0.92	negative	-0.90
s3	0.71	0.24	0.05	positive	+0.66
s4	0.02	0.96	0.02	neutral	0.00
s5	0.11	0.85	0.04	neutral	+0.07

- Read s3 first: the dictionary called it strongly negative, and here p(positive) is 0.71.
- Read s4 second: 0.96 on boilerplate is correct, and it will dilute every document score.



Step 4: Four Aggregation Rules on the Same Sentences

The model is fixed. Only the arithmetic changes, and the answer changes with it.

Rule	Calculation	Result
Net count	$(2 - 1) / 5$	+0.200
Drop neutrals, then net	$(2 - 1) / 3$	+0.333
Mean signed probability	$(0.93 - 0.90 + 0.66 + 0.00 + 0.07) / 5$	+0.152
Dispersion of the signed scores	$\text{sqrt}(2.000 / 4)$	0.707

- Three defensible rules, answers a factor of two apart, and nobody is wrong.



The Aggregation Decision

Each rule assumes something, and each one fails somewhere different.

Rule	What it assumes	Where it fails
Net count	Every sentence counts equally	Boilerplate dilutes a strong signal, as s4 did
Drop neutrals, then net	Neutral means no content	Loses the fact that the filing was quiet
Mean signed probability	Confidence carries information	Miscalibration propagates into the feature
Dispersion	Disagreement is the signal	Unstable on a three-sentence document

- Report at least two rules, and show that your conclusion does not depend on the choice.
- A middle path: length-weight the sentences, so a heading fragment counts for less.



Step 5: The Lag Rule and the Sign Change

Four filings in one week, and two ways of assigning them to days.

Filing	Acceptance time (ET)	Score	Correct assignment	Naive assignment
8-K #1	Mon 09-08 07:12	+0.20	Mon 09-08, pre-open	Mon 09-08
8-K #3	Wed 09-10 16:42	-0.55	Thu 09-11, after the close	Wed 09-10
8-K #4	Thu 09-11 13:20	+0.11	Thu 09-11	Thu 09-11

Daily feature	Monday	Wednesday	Thursday
Correct rule	+0.20	-0.29	$(-0.55 + 0.11) / 2 = -0.22$
Naive rule	+0.20	$(-0.29 - 0.55) / 2 = -0.42$	+0.11

- Wednesday changes sign under the naive rule, on the same documents and the same model.



The Resulting Sentiment Series

The daily series the pipeline produces, for one firm over one week.

Date	Filings	Sentences scored	Daily score	Feature value used
Mon 09-08	1	5	+0.20	+0.20
Tue 09-09	0	0	undefined	0.00 or carried forward
Wed 09-10	1	22	-0.29	-0.29
Thu 09-11	2	31	-0.22	-0.22
Fri 09-12	0	0	undefined	0.00 or carried forward

- The two blank days are the hardest column, and all three ways of filling them differ.
- Counts of 5, 22 and 31: a score from five sentences is far noisier than one from thirty-one.



Entity-Level Sentiment

"Apple is cutting orders for the modem, moving the work in-house."

- Good for one firm, bad for its supplier, so the sentence average comes out neutral.
- The task you want: one label per company mentioned, not one per sentence.
- Why it matters: filings and news name counterparties on almost every page.



Tone Against Polarity

The two words are used interchangeably in student reports, and they are not the same measurement.

	Polarity, or sentiment	Tone
The question asked	Is this text good or bad news?	How positively is the writer speaking?
What it is about	The event the text reports	The language the writer chose
Usual measurement	A classifier label per sentence	A ratio of positive to negative words in a document
Where it goes wrong	One label hides four economic meanings	A firm can write warmly about bad news

- Tone is polarity measured over a whole document and attributed to the writer, not the event.



Measurable Quantities in Text Beyond Tone

Tone is one of several quantities you can extract from text.

Quantity	What it measures	How it is usually computed
Attention	How much is written about the firm at all	Documents or posts per day
Uncertainty	How firmly the writer commits	Hedging-term frequency, or a classifier
Novelty	Whether today's document says anything new	Similarity to the firm's recent documents
Dispersion/Disagreement	Whether the writers agree with one another	Dispersion of scores within a day
Named events	What kind of thing happened	Item codes, or extraction into a schema

- The frame is wider than polarity, and the other rows are often easier to measure well.



The Same Filing, Read Five Ways

One 8-K, five quantities, and five different claims about it.

Quantity	Value for this filing	What it would let you say
Tone	Mean signed score +0.15	Mildly positive on average
Attention	Fourth filing this month, median one	Something unusual is happening
Uncertainty	Hedging at three times the firm's own rate	Management is committing to less
Novelty	Cosine 0.93 to last month's filing	Most of the text is recycled
Named event	Item 2.02, results of operations	An earnings release, not a notice

- Tone and novelty are in tension here, and deciding which to trust means computing both.



Social and News Text as a Second Corpus

Free, fast, enormous, and much harder to trust than a filing.

	SEC filings	News wire copy	Social and forum posts
Volume per firm per day	0 on most days	0 to a few dozen	Hundreds to millions
Register	Formal, legal, repetitive	Edited, factual	Slang, sarcasm, tickers
Timestamp quality	Acceptance minute, exact	Publication time	Blurred by reposts
Main failure mode	Boilerplate dilutes the score	Near-duplicates	Bots and manipulation

- A few trustworthy documents against many untrustworthy ones, and the choice is a research design.



Message Volume Against Message Tone

Antweiler and Frank (2004), Is All That Talk Just Noise?, Journal of Finance 59(3).

Over 1.5 million messages about 45 companies, each classified buy, hold or sell.

Measure built from the messages	Predicts returns?	Predicts volatility?
Message volume	Weakly	Yes, robustly
Bullishness, the average tone	Barely, and tiny	Less than volume does
Disagreement among posters	No	Yes; dispersion predicts movement

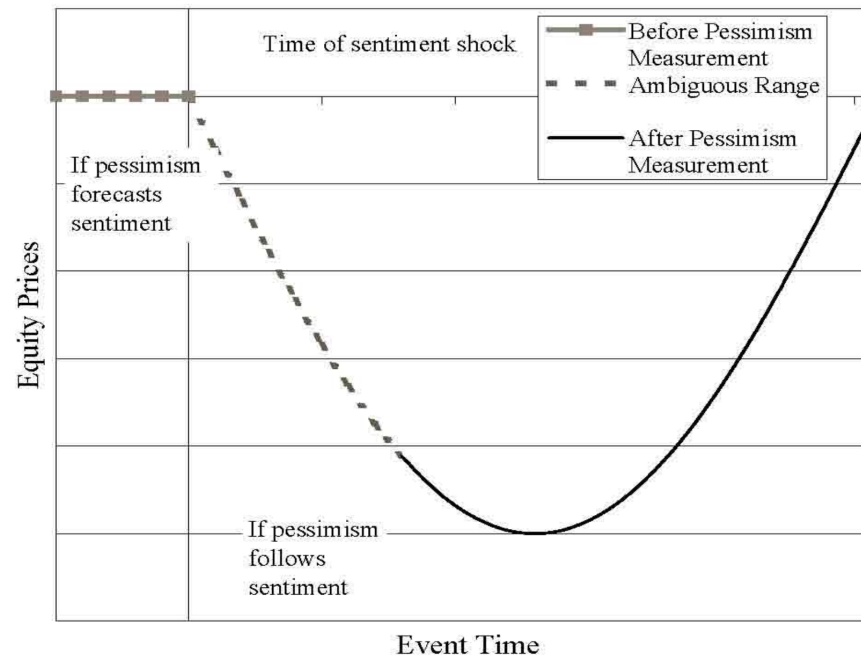
- People post when something happens, and something happening moves prices.



Media Pessimism and Return Reversal

Tetlock (2007), Giving Content to Investor Sentiment, Journal of Finance 62(3).

- The measure: one daily pessimism score from the language of a widely read news column.
- The finding, in three parts: pressure downward, reversion within days, and high volume.
- Read the reversal carefully: this forecasts temporary pressure, not value.



Text as a Predictor of Risk Rather Than Direction

Kogan et al. (2009), Predicting Risk from Financial Reports with Regression, NAACL-HLT.

Unigram and bigram counts over about 60,000 annual reports, predicting realized volatility.

Predictor	Result
Historical volatility before the filing	The obvious baseline, and a strong one
Text alone	Comparable to the historical baseline
Text plus historical volatility	Better than either alone
The same text, for return direction	Nothing usable

- Text says how much a price will move even when it cannot say which way.

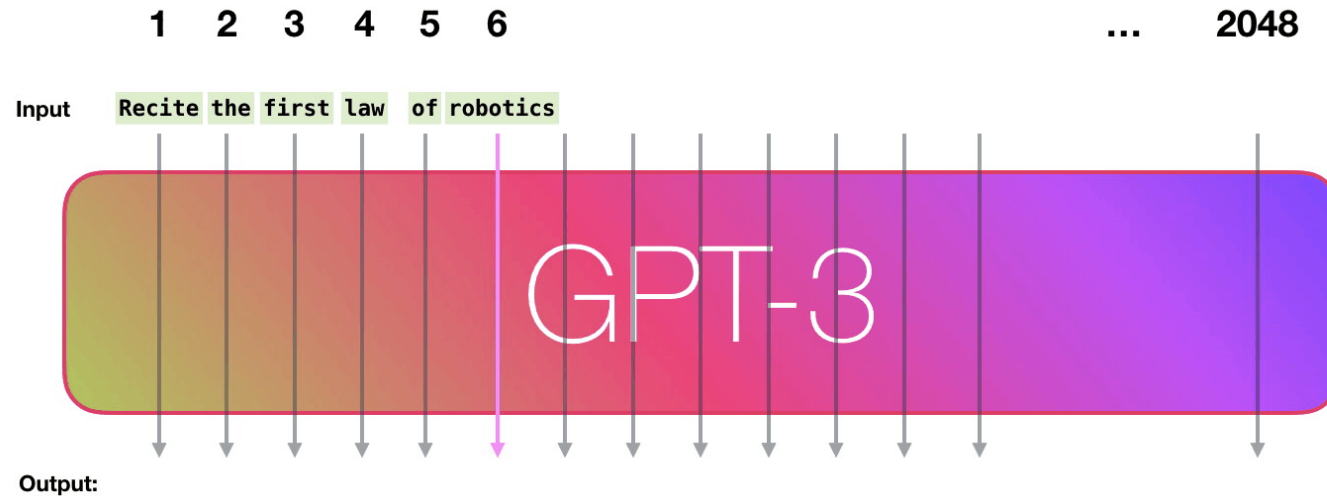


Decoders and Next-Token Prediction

A third generation of models, and it is trained on a different question.

- A decoder reads only the text to its left and returns a probability distribution over the next token.
- The training objective is that single question, asked at every position of an enormous unlabeled corpus.
- BERT sees both sides and fills a hole; a decoder sees one side and continues the text.
- Neither objective is more correct: one is built to represent, the other is built to produce.

Encoder or decoder is a statement about which words the model is allowed to look at.



From Next-Token Pretraining to Following Instructions

A model that only continues text is not yet a model that answers you.

- Pretraining on next-token prediction gives fluency and a great deal of stored knowledge.
- A second stage trains the same model on written instructions paired with acceptable responses.
- The result is a model that treats your input as a task to carry out rather than a prefix to continue.
- Claude and GPT are instruction-following models of exactly this kind, at a much larger scale.

Sentiment then needs no training run of your own: the task can be stated in a sentence.



Zero-Shot Prompting for Sentence Sentiment

No training and no labels. Write the task in words, read the answer.

For the sentence below, from an SEC 8-K exhibit, answer **positive, negative or neutral**, with one word and nothing else.

Sentence: The Company reversed the impairment charge.

- Zero-shot means the instruction alone; few-shot adds a handful of labeled examples inside the same prompt.
- Lopez-Lira and Tang (2023), arXiv:2304.07619: headlines scored this way predicted returns.
- Kim, Muhn and Nikolaev (2024), arXiv:2407.17866: beating the median analyst.

The same prompt also returns an entity-level label, with no extra step.



Fine-Tuned Encoder Against Zero-Shot Prompting

Both are in use, and the choice is made on cost and reproducibility rather than novelty.

Property	Fine-tuned encoder	Instruction-following LLM, prompted
Cost per document	Zero after the download	Priced per token
Throughput	Hundreds of sentences a second	Network-bound and rate-limited
Reproducibility	Bit-identical on rerun	Non-deterministic and silently versioned
A new label scheme	Collect labels and fine-tune	Write it into the instruction
Numbers and entities	Weak	Handled, with an explanation attached

- A million sentences run through an encoder on one notebook GPU in under two hours.



The Three Generations on the Same Sentence

The same sentence, s3, read once by each of the three methods.

- s3: The Company reversed \$42 million of the impairment charge recorded in the prior year.

Generation	Verdict on s3	What it could and could not see
Dictionary	Strongly negative	Saw impairment and charge. Could not see reversed
Domain-adapted encoder	positive, $p = 0.71$	Saw that a reversal adds income. Not the \$42 million
Instruction-following LLM, zero-shot	positive, with an explanation	Reads the amount. The explanation is not a log

- Each generation fixed the previous one's specific failure, and none made the earlier ones useless.



Failure Modes Specific to Social Text

Five problems that do not arise on filings.

Problem	What it does to your feature	The practical response
Duplicates and reposts	One event counted fifty times	Deduplicate on near-identical text first
Bots and coordinated posting	Volume reflecting a campaign	Drop accounts by posting rate
Sarcasm and irony	The sign is wrong on the interesting posts	Prefer volume and dispersion measures
Ticker ambiguity	Cashtags collide with common words	Require the cashtag, hand-check a sample
Survivorship in the archive	Deletion is not random	State it; you cannot repair it

- The first two inflate a count, and a count is the feature most likely to work.



Timestamps and the Alignment Rule

The alignment discipline, stated once for both registers.

	SEC filing	Social post
What the timestamp records	The minute the SEC accepted it	The minute the platform recorded it
Is it the event time?	No; the rule allows four business days	A repost carries a later time
Time zone	Eastern, stated	Often UTC, often unlabeled in an export
Editable afterward?	No; an amendment is a new filing	Yes on some platforms, invisibly

- Convert to Eastern, then move anything from 16:00 onward to the next session.

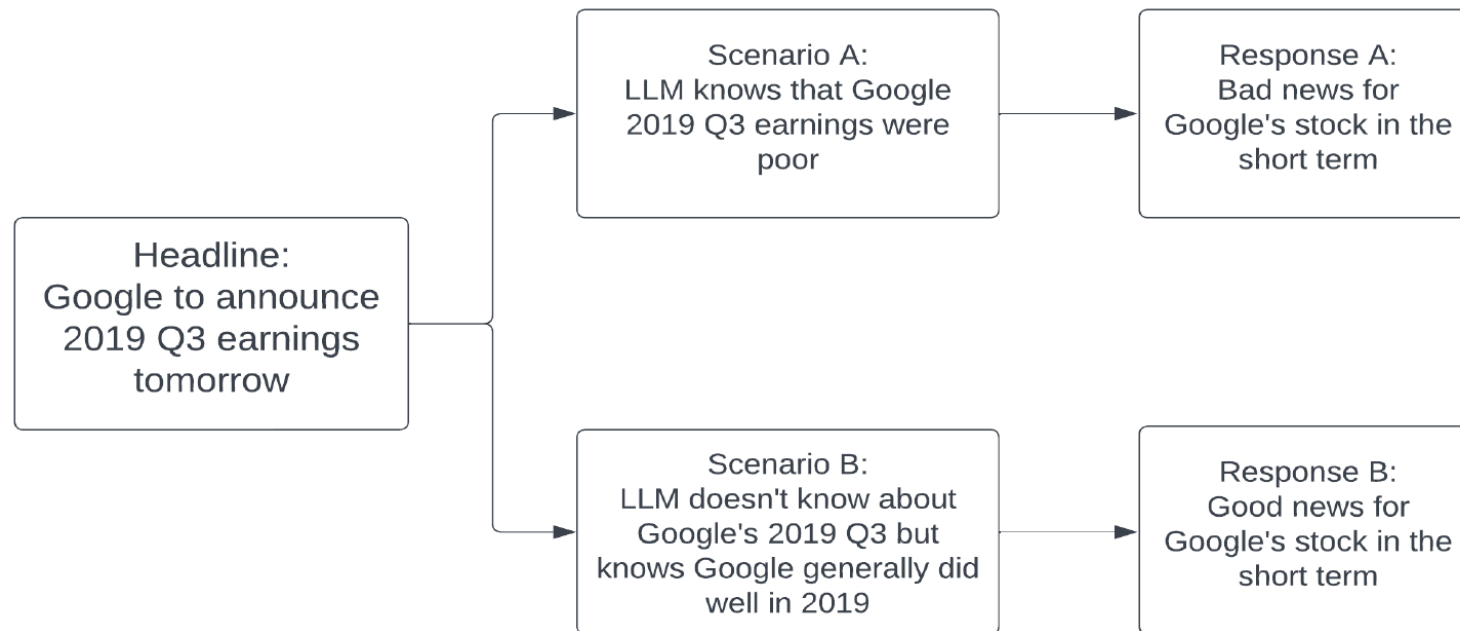
An unlabeled time zone is a fourteen-hour look-ahead waiting to happen.



Contamination and Look-Ahead

The failure mode that no leakage check on your own data will catch.

- Your splitting is correct, and the model still knows the answer.
- The route is the weights: the corpus was assembled after the event you predict.
- Worst for the largest firms, which are the names in the S&P 500.



Measuring Contamination by Scoring Everything Twice

Assessing Look-Ahead Bias in Stock Return Predictions Generated By GPT Sentiment Analysis, arXiv:2309.17322.

Score each document as written, then again with the name, tickers and dates masked.

Version scored	Annualized long-short return	Share of the original effect
Headlines as written	6.2%	100%
Company name and dates masked	3.4%	55%
The gap, attributable to recognition	2.8 percentage points	45%

- These magnitudes are a worked template, not the paper's own numbers.



Four Mitigations for Contamination

Four responses to contamination, in order of how much they help.

1. Anonymize before scoring: strip names, tickers and dates.
2. Report the knowledge cutoff as a property of the instrument.
3. Run the placebo: if the model can date the text, it remembers it.
4. Prefer data published after the cutoff.

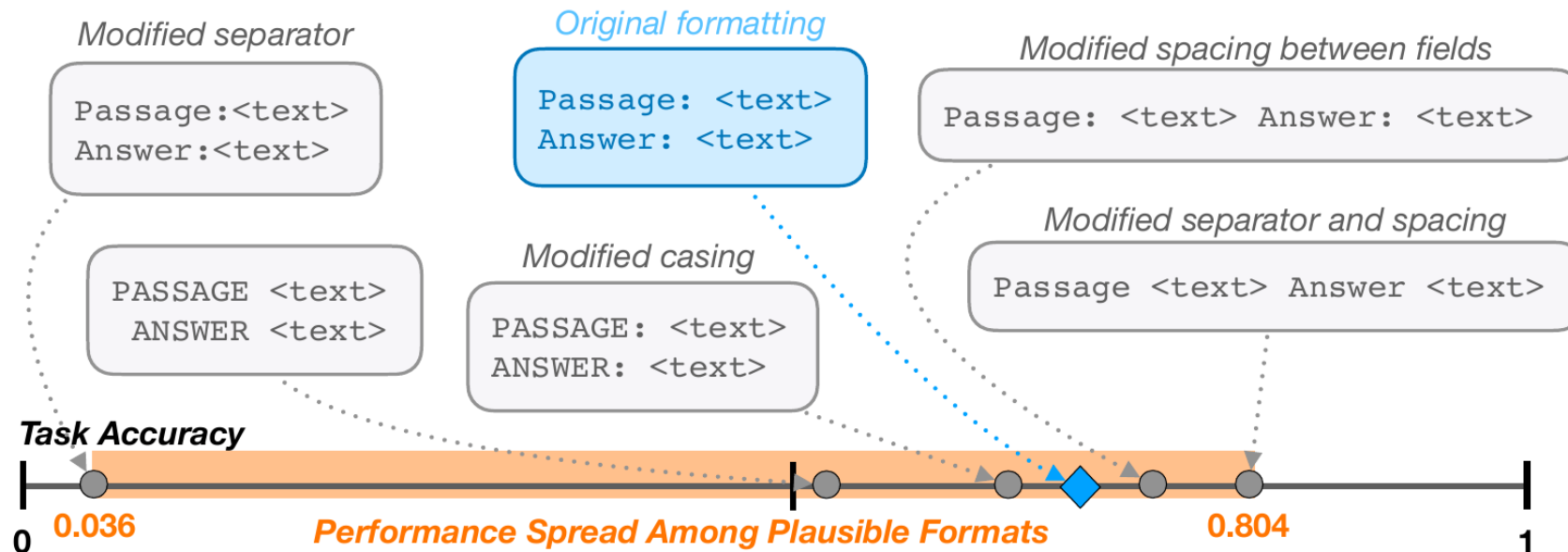
Score everything twice. If the two scores differ, the gap came from memory.



Prompt Sensitivity

Sclar et al. (2024), Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design, ICLR.

- Cosmetic changes move accuracy: a colon for a dash, the options reordered.
- Not a bug that gets fixed: it is a property of the instrument.
- Report the score under several paraphrases, not under your favorite one.



Arrival Count as the Baseline

Movement responds to the arrival rate of information. Direction is the hard part.

- A count needs nothing: no classifier, no prompt, no rule, and it cannot leak.
- A polarity score adds measurement error to a variable in a regression.
- Everything today was error: two lag rules, a masking gap, prompt spread.

